



A Geospatial Vertical Event **GEOIMPATHON 1.0**

GEOSPATIAL IDEAS FOR GREENER EARTH

30.09.2026

General Instructions

- Select any one problem statement from any of the following task domains
 - Urban Green Space Mapping
 - Water Resource Management
 - Agricultural Sustainability
 - Disaster Risk Analysis
- Each team should develop:
 - Satellite Data → Image Processing → Geospatial Analysis → Indicator/Model → Interactive Map/Dashboard → Decision / Recommendation
- Teams can use:
 - Google Earth Engine
 - [Sentinel-1 SAR](#)
 - [Sentinel-2 MSI](#)
 - [Landsat 5/7/8/9](#)
 - [SRTM/DEM](#)
 - MODIS/VIIRS
 - CHIRPS rainfall
 - ERA5 climate data
 - OpenStreetMap
 - Google Earth Engine Apps
 - Python/QGIS

Recommendations for Datasets

- For a 2-day image-processing hackathon, I would restrict participants mainly to these because they are accessible directly in Google Earth Engine and don't require downloading external datasets:
 - Optical
 - Sentinel-2
 - Landsat 5/7/8/9
 - MODIS
 - SAR
 - Sentinel-1
 - Terrain

- SRTM
 - NASADEM
 - Copernicus DEM
- Land cover
 - Dynamic World
 - ESA World Cover
- Dynamic World is especially useful because it provides 10 m land-cover information from 2015 onward, with Sentinel-2-based observations and frequent revisits.
 - Rainfall/climate
 - CHIRPS
 - GPM
 - ERA5
 - ERA5-Land
- CHIRPS is particularly suitable for your hackathon because it provides a long rainfall record from 1981 to near-present, including daily products.
 - Water
 - JRC Global Surface Water
 - Sentinel-1
 - Sentinel-2
 - Dynamic World
 - Vegetation
 - Sentinel-2 NDVI/EVI
 - Landsat NDVI
 - MODIS vegetation indices
 - Disaster
 - Sentinel-1 flood mapping
 - Sentinel-2 damage/change mapping
 - CHIRPS/GPM rainfall
 - DEM-derived slope/elevation
 - MODIS/VIIRS fire

Submission Instructions

- Teams must complete the challenge within a strict **6-hour window**. **The Team Lead must submit the following deliverables through the designated Google Form prior to the deadline:**
 - **Source Code Repository:** A link to a public GitHub repository under the Team Lead's account containing the complete, committed project code.
 - **Live Deployment:** A functional URL pointing to the deployed project, accessible as a live web application or dashboard.
 - **Video Demonstration:** A public or unlisted YouTube link featuring a concise demonstration of the working project (2 to 3 minutes in duration).

Problem Statements

Domain 1: Urban Green Space Mapping	
1.1	Urban Green Space Accessibility Mapping Challenge: Identify areas where residents have poor access to urban green spaces. Tasks: • Map vegetation/green spaces using Sentinel-2. • Calculate NDVI. • Extract parks/green areas. • Create buffers around green spaces. • Identify residential areas outside reasonable walking/access zones. Expected output: Urban Green Accessibility Index + priority areas for new green spaces.
1.2	Urban Heat Island and Green Cover Relationship Challenge: Identify urban heat hotspots and investigate whether lack of vegetation contributes to higher land surface temperature. Tasks: • Generate LST using Landsat. • Calculate NDVI. • Identify urban heat hotspots. • Analyze relationship between NDVI and LST. Expected output: Urban Heat Island map + vegetation–temperature relationship + hotspot map
1.3	Green Corridor Identification Challenge: Design potential green corridors connecting fragmented urban vegetation. Tasks: • Extract vegetation patches. • Calculate vegetation density. • Identify fragmented green spaces. • Develop a connectivity/cost-surface approach. • Identify potential ecological corridors. Expected output: Existing green-space network + proposed green corridors.
1.4	Urban Tree/Vegetation Loss Detection Challenge: Detect significant loss of urban vegetation over the last 10–20 years. Tasks: • Compare multi-temporal Landsat/Sentinel imagery. • Generate NDVI for different years. • Detect vegetation-loss hotspots. • Quantify changes by administrative unit. Expected output: Vegetation-loss map + statistics + priority conservation locations.

Domain 2: Water Resource Management	
2.1	Surface Water Change Detection Challenge: Monitor the expansion and contraction of lakes, ponds and reservoirs. Tasks: • Extract water using NDWI/MNDWI. • Compare multiple years. • Calculate water-area changes. • Identify water-loss/gain hotspots. Expected output: Multi-temporal water-body change map and statistics.
2.2	Urban Waterbody Encroachment Detection Challenge: Identify water bodies experiencing reduction in area due to urban development. Tasks: • Extract historical water extent. • Extract recent water extent. • Overlay built-up expansion. • Quantify waterbody area loss. Expected output: Waterbody encroachment-risk map.
2.3	Watershed-Based Water Resource Priority Mapping Challenge: Identify micro-watersheds requiring water conservation interventions. Parameters: • DEM • Slope • Drainage density • Rainfall • NDVI • Land use • Soil • Surface-water occurrence Expected output: Water conservation priority zones.
2.4	Rainfall–Surface Water Response Analysis Challenge: Investigate how rainfall variability affects surface-water availability. Tasks: • Obtain rainfall time series. • Extract surface-water extent. • Compare rainfall and water-area changes. • Identify drought-sensitive water bodies. Expected output: Rainfall–water response analysis and vulnerable waterbody map.

Domain 3: Agricultural Sustainability	
3.1	Crop Health Monitoring System Challenge: Develop a satellite-based crop health monitoring system. Tasks: • Calculate NDVI/NDRE or other vegetation indices. • Generate crop-health classes. • Compare crop condition through time. • Identify stressed agricultural fields. Expected output: Healthy/moderate/stressed crop map.
3.2	Agricultural Drought Stress Detection Challenge: Identify agricultural areas experiencing vegetation stress due to insufficient moisture. Possible indicators: • NDVI • NDMI • LST • Rainfall • Soil moisture Expected output: Agricultural drought-stress map.
3.3	Cropping Intensity Mapping Challenge: Identify single-, double- and multiple-cropped areas using satellite time-series data. Tasks: • Generate monthly/seasonal NDVI profiles. • Detect crop-growth cycles. • Classify agricultural intensity. Expected output: Cropping-intensity map and seasonal statistics.
3.4	Irrigated vs Rain-Fed Agriculture Mapping Challenge: Develop a remote-sensing approach to distinguish irrigated and rain-fed agricultural areas. Tasks: • Analyze vegetation seasonality. • Use NDVI/NDMI. • Integrate rainfall information. • Identify persistent vegetation during dry periods. Expected output: Irrigated/rain-fed agricultural classification.
Domain 4: Disaster Risk Analysis	
4.1	Rapid Flood Mapping Using Sentinel-1 Challenge: Develop an automated system for rapid flood detection.

	Tasks: • Select pre-flood and post-flood Sentinel-1 images. • Perform SAR processing. • Detect water expansion. • Calculate affected area. Expected output: Before/after flood map + affected settlements/roads/agricultural land.
4.2	Landslide Susceptibility Mapping Challenge: Identify areas susceptible to landslides using terrain and environmental factors. Parameters: • Slope • Elevation • Aspect • rainfall • NDVI • drainage density • land use • geology, if available Expected output: Landslide susceptibility map with 3–5 classes.
4.3	Wildfire Burn-Area Assessment Challenge: Detect and quantify vegetation affected by wildfire. Tasks: • Use Sentinel-2/Landsat. • Calculate NBR/dNBR. • Map burned areas. • Estimate burn severity. Expected output: Burned-area and burn-severity map.
4.4	Disaster Exposure Mapping Challenge: Identify infrastructure and settlements exposed to a selected hazard. Workflow: Hazard layer + Roads + Buildings + Population + Land use → Exposure analysis Expected output: Hazard-exposure map and affected infrastructure statistics.